
Mini Project: Fourier Series Signal Reconstruction

Goal

Understand Fourier series by reconstructing periodic functions using partial sums and studying how they converge.

Task 1 — Choose your signals

Work with at least **three periodic functions**:

- Square wave
- Sawtooth wave
- Triangle wave

Make sure each is defined on a finite interval (e.g. $([-\pi, \pi])$) and extended periodically.

Task 2 — Write Fourier series formulas

For each function:

- Write down its known Fourier series representation
 - Identify whether it uses:
 - sine terms only (odd symmetry)
 - cosine terms only (even symmetry)
 - both
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Task 3 — Build partial sums

For each signal:

- Construct partial Fourier series approximations using:
 - $N = 1$
 - $N = 3$
 - $N = 5$
 - $N = 10$
 - $N = 50$ (or higher if possible)
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Task 4 — Visualization

For each signal:

- Plot the original function
 - Plot each partial sum on the same axes
 - Observe convergence behavior
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Task 5 — Analyze convergence

For each function, describe:

- How quickly the approximation improves as N increases
 - Whether oscillations appear near discontinuities
 - Whether smooth functions converge faster than discontinuous ones
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Task 6 — Gibbs phenomenon

Specifically investigate:

- Does overshoot occur near discontinuities?
 - Does the overshoot disappear or persist as N increases?
 - Is the width of the oscillation region shrinking or constant?
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Task 7 — Comparison across functions

Compare all three signals:

- Which one is easiest to approximate?
 - Which one needs the most terms?
 - How does smoothness affect Fourier convergence?
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Task 8 — Summary interpretation

Write a short conclusion covering:

- What Fourier series is doing physically
 - Why it works as a decomposition into oscillatory modes
 - What you learned about frequency content and convergence
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Optional extension (if you want more depth)

- Try a fourth function of your choice
 - Experiment with removing odd/even symmetry constraints
 - Estimate error between approximation and true function
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